CTGDiff: A Conditional Diffusion Model for Cardiotocography Signal Synthesis

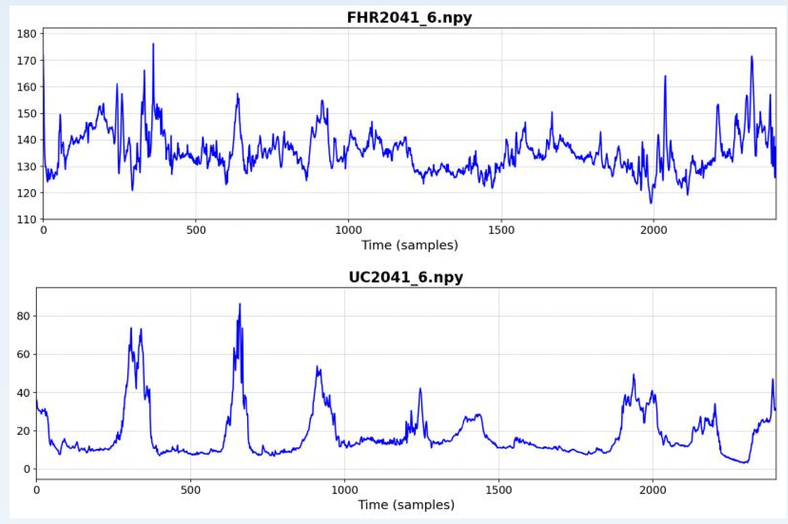
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Introduction

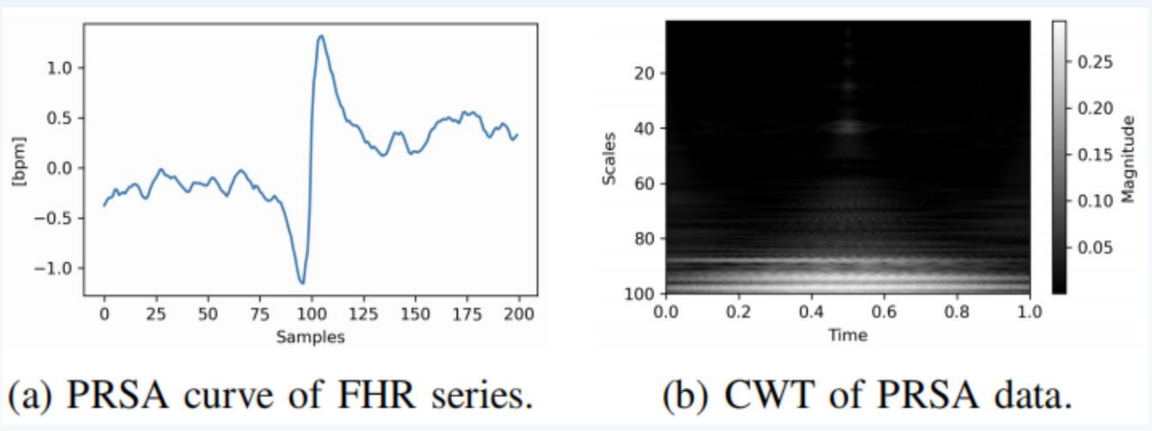
- In this work, we present CTGDiff, a novel conditional diffusion model designed for generating synthetic Fetal Heart Rate (FHR) and Uterine Contraction (UC) signals.
- Introducing the PRSA curve for generating paired spectrogram conditions.
- We conduct a thorough comparison with several state-of-the-art GAN models tailored for FHR signal generation.

Synthetic CTG signal



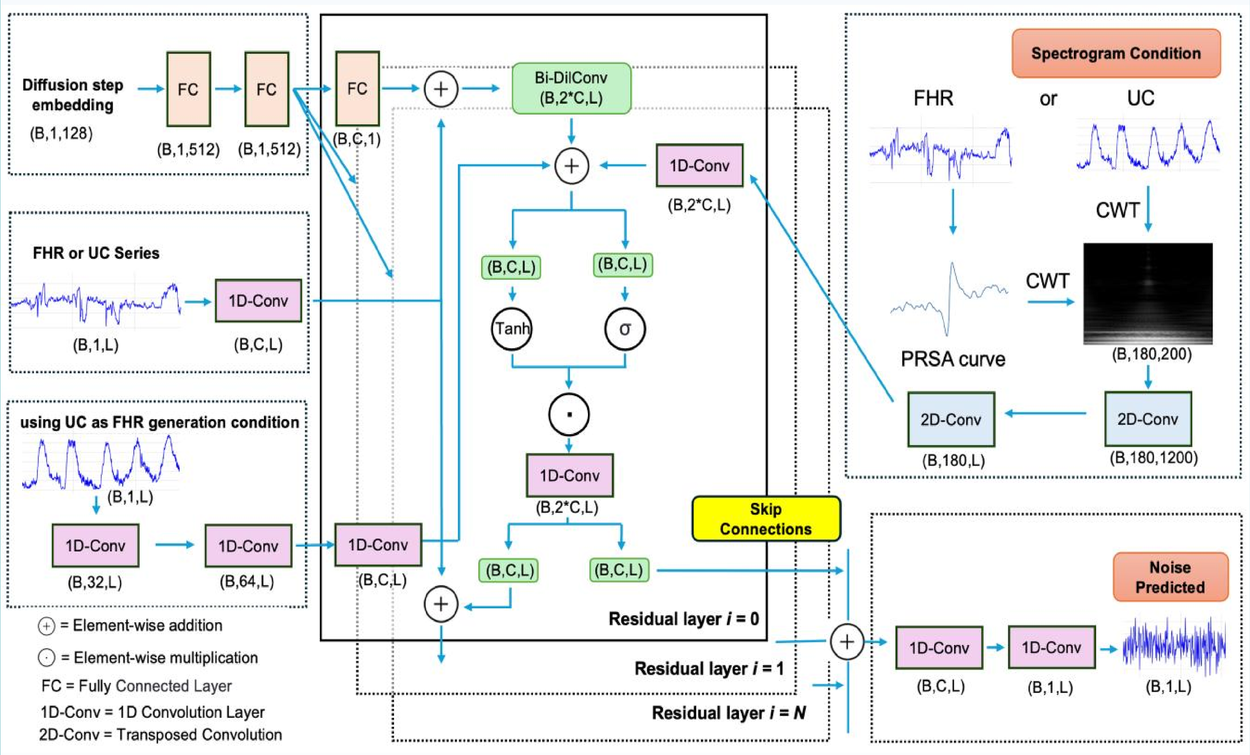
PRSA Curve and Spectrogram

- Using the Phase-Rectified Signal Averaging (PRSA) spectrogram as the label for each signal.
- This method enhances the model's ability to distinguish between normal and pathological features in raw CTG data.



(a) PRSA curve of FHR series. (b) CWT of PRSA data.

Model Architecture



- Our model is designed to jointly train and generate multivariate signals of FHR and UC within a mixed training framework.
- We utilized the CTU-CHB Intrapartum Cardiotocography dataset and segment each signal into 20-minute lengths for training.
- For FHR generation, the previously synthetic UC signal serves as the first condition, while the FHR spectrogram (transformed from PRSA and CWT) is used as the second condition.

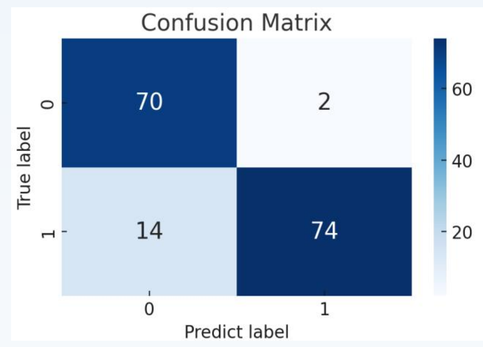
Experiment

- We trained a model on both real and synthetic signals to distinguish normal from abnormal FHR signals, and tested its accuracy on real signals.
- We included evaluations of 160 synthetic samples by three obstetricians

TABLE I: Comparison of Different Models for Synthesis

Model	Evaluation Metrics				
	RE(↓)	FD(↓)	MMD(↓)	SWD(↓)	PRD(↓)
SeqGAN [28]	10.324	0.974	6.527	0.541	10.635
DCGAN [29]	9.532	0.795	5.267	0.254	10.431
CGAN [30]	9.012	0.763	5.022	0.254	9.431
CWGAN [31]	8.523	0.715	4.568	0.173	0.702
FHRGAN [18]	8.059	0.682	1.975	0.086	5.437
CTGGAN [19]	5.621	0.614	0.215	**0.012**	4.821
DDPM [25]	7.059	0.682	6.323	0.521	9.786
CTGDiff	**3.951**	**0.291**	**0.004**	0.245	**0.646**

(a) evaluation results



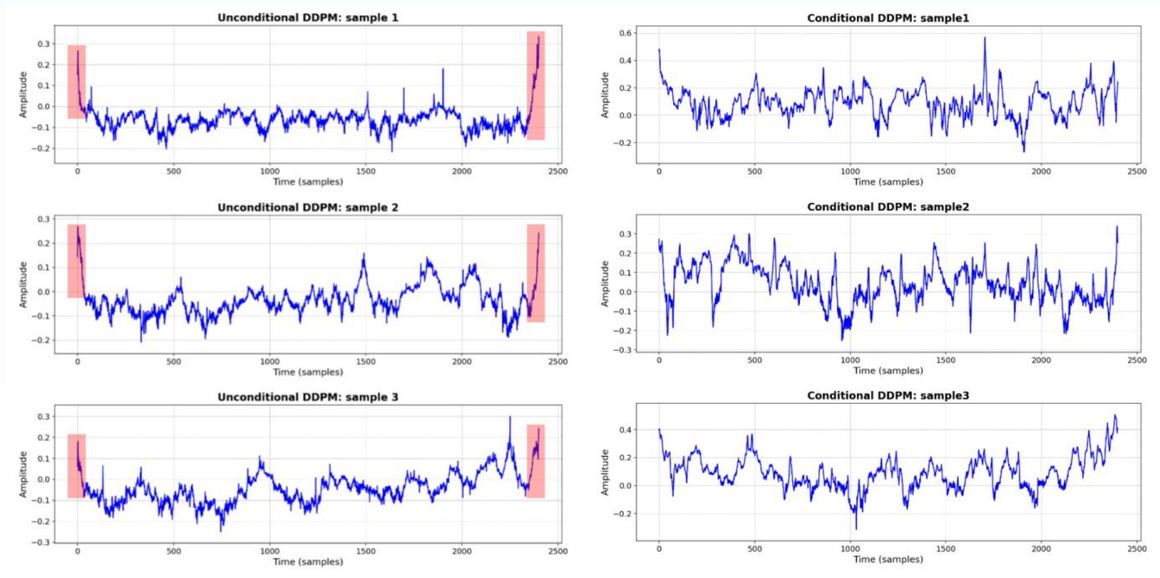
(b) Confusion Matrix for Manually Classification by An Obstetrician.

- We compared the fidelity of the synthetic data using various evaluation metrics.

No.	Real Signal	Synthetic Signal	Accuracy(↑)
1	100%	0%	75.0%
2	90%	10%	73.0%
3	80%	20%	73.0%
4	70%	30%	69.0%
5	60%	40%	67.1%
6	50%	50%	68.5%

(c) Classification Accuracy with Different Ratios of Real and Synthetic CTG Signals

- We compared the signal quality of conditional and unconditional generation.



(d) It is evident that the data generated by unconditional DDPM exhibit similar patterns at the beginning and end of the signals

References

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